PROJECTION FORMULA FOR AFFINE MORPHISMS AGAINST F_*O_X

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof constructs the comparison map via the counit of the pull-push adjunction, reduces etale locally to the affine scheme case, and quotes the scheme projection formula

Editorial note. The label lemma-affine-morphism-pull-push names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- vanishing of higher direct images under affine morphisms, so f_*O_X -> Rf_*O_X is an isomorphism (Cohomology of Spaces, Lemma spaces-cohomology-lemma-affine-vanishing-higher-direct-images)
- the canonical projection map E tensor^L Rf_*O_X -> Rf_* Lf^* E adjoint to the counit Lf^*Rf_*O_X -> O_X
- the comparison of D_QCoh on the etale site with schemes (lemma-derived-quasi-coherent-small-etale-site of the same chapter) and the compatibility of total direct images (remark-match-total-direct-images of the same chapter)
- the scheme projection formula for affine morphisms (Derived Categories of Schemes, Lemma perfect-lemma-affine-morphism-pull-push)

Source reference. Original reference 071Q: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 402–402. Prerequisite pointer invoked at proof line(s) 1011; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08I9: sources/stacks-chow/tags.txt, source lines 10807–10807. Prerequisite pointer invoked at proof line(s) 1010; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08GH: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 896–896. Prerequisite pointer invoked at proof line(s) 1012; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 073H: sources/stacks-chow/tags.txt, source lines 9061–9061. Prerequisite pointer invoked at proof line(s) 994; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be an affine morphism of algebraic spaces over S. For E in $D_{QCoh}(\mathcal{O}_Y)$ we have $Rf_*Lf^*E = E \otimes^{\mathbf{L}}_{\mathcal{O}_Y} f_*\mathcal{O}_X$.*

Proof. Since f is affine the map $f_*\mathcal{O}_X \to Rf_*\mathcal{O}_X$ is an isomorphism (Cohomology of Spaces, Lemma 073H). There is a canonical map $E \otimes^{\mathbf{L}} f_*\mathcal{O}_X = E \otimes^{\mathbf{L}} Rf_*\mathcal{O}_X \to Rf_*Lf^*E$ adjoint to the map

$$Lf^*(E \otimes^{\mathbf{L}} Rf_*\mathcal{O}_X) = Lf^*E \otimes^{\mathbf{L}} Lf^*Rf_*\mathcal{O}_X \longrightarrow Lf^*E \otimes^{\mathbf{L}} \mathcal{O}_X = Lf^*E$$

coming from $1 : Lf^*E \to Lf^*E$ and the canonical map $Lf^*Rf_*\mathcal{O}_X \to \mathcal{O}_X$. To check the map so constructed is an isomorphism we may work locally on Y. Hence we may assume Y and therefore X is affine. In this case the problem reduces to the case of schemes (Derived Categories of Schemes, Lemma 08I9) via Lemma 071Q and Remark 08GH. □